ISLAND DEV CREW CONSULTING · GENERAL CONTRACTING & AI SYSTEMS

Oakwood University AI Studio & Engineering Lab

Turnkey proposal & execute-ready estimate for a 10-seat AI production studio, a hands-on engineering/soldering makerspace, and the department-wide power, network, camera, and access-control infrastructure that protects the investment — orchestrated end-to-end by Island Development Crew.

Prepared: June 2026
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What this is. A planning-grade, audit-aware General-Contractor proposal built from an on-site walkthrough (May 26, 2026), 16 verified site photographs, and current June-2026 vendor pricing with named Huntsville subcontractors. IDC orchestrates every trade under one accountable point of contact, one transparent fee, and one progress dashboard. *Figures are planning ranges; each line is re-quoted to a sealed vendor/sub bid before purchase.*

10-Seat AI Studio · Rm 409 | Soldering/Math Lab · Rm 407 | Engineering · Rm 305 + 310 | UniFi + pfSense Network | UniFi Protect Cameras | Multi-Door Access | ≥4 Bambu Lab Printers | 2084 / Lennox Curriculum

RECOMMENDED EXECUTION — ALL-IN (IDC-ORCHESTRATED)

$172,000

Range $170,625 – $198,498. Includes equipment, subbed facility work, GC coordination, contingency, and the 10% IDC orchestration fee. Title III grant-eligible portion ≈ $93.5K; facility/capital portion ≈ $60.1K.

Mobilization deposit initiates procurement & sub scheduling (see Payment Schedule, p.7).

Secure payment: **Island Dev Crew Stripe payment link**
(insert IDC Stripe link / Payment Link)

Signing: signature-ready PDF with direct blocks (p.7); DocuSign / SignWell / Dropbox Sign accepted.

Department-Wide Vision (full robust build incl. dept cabling, dept cameras, Room-405 bathroom & raised floor) is presented on p.10 as the donor / naming-gift envelope at **$245K–$330K**.

The opportunity

Oakwood does not need another computer lab. It needs an **AI production studio** where Math & Computer Science students *ship* AI — apps, agents, fabricated hardware, and research artifacts — backed by reliable power, network, and security. As the only Seventh-day Adventist HBCU, Oakwood can also give students a credible **responsible-AI ethics** voice (curriculum, p.9).

Why IDC as GC

One orchestrator manages every vendor and Huntsville subcontractor (electrical, low-voltage, access, GC), holds a single **3–7% coordination markup** on subbed facility work and a transparent **10% orchestration fee**, and keeps faculty & administration informed through a live progress dashboard with weekly/monthly reports.

1 · Executive Summary & How to Read This

This package separates the project into the three quote families requested, because each draws on **different funding**. Selling all three as one grant line will be clawed back at audit.

Track	Scope	Planning range	Funding lane
Quote 1 — AI Studio + Engineering	Compute, Bambu fleet, soldering/CNC, AV + 3 boards, furniture, consumables, software	$88.2K–$98.8K	Mostly **Title III grant-eligible**
Quote 2 — Power / Electrical	Subpanel, dedicated circuits, raceway, raised floor, permits	$54.4K–$61.0K	**Facility — NOT grant** (capital / donor)
Quote 3 — Network + Cameras + Access	UniFi + pfSense, Protect cameras, 4-door Access; + structured cabling labor	Equip in Q1·Q3; cabling facility	**Mixed** (electronics grant / cabling facility)

2 · Two Tiers — Recommended vs. Vision

Recommended Execution — ~$172,000 all-in

Studio/lean compute, lab-focused facility scope (power + cabling to the labs + MDF; raised floor in the AI Studio only; *no* Room-405 bathroom). This is the number to commit to — it sits at the top of the $150–175K band you targeted and is what the execute-ready invoice (p.7) bills.

Sub-$170K lever: a single-lab camera pilot or 2-door access pulls the point toward $160–165K (your call).

Department-Wide Vision — $245K–$330K

The full robust transformation: pooled compute (RTX PRO 6000), dept-wide cabling (80 drops + fiber), dept-wide cameras, full electrical, raised floor *plus* the Room-405 bathroom and dept room-mod, 5-door access. This is the **donor / naming-gift** envelope (p.10) — not an execution commitment today.

3 · The Funding Firewall (critical)

Title III covers **supplies (<$2,000/unit)** and **equipment (≥$2,000/unit)** tied to the academic mission, with an audit trail and a **June 30** spend pressure. It does **not** cover **facilities** — electrical service, structural cabling labor, raised floor, construction, HVAC, the bathroom. Quote 2 in full and the cabling-labor / door-install portions of Quote 3 must come from **capital, physical-plant, donor, or match funds**. (The grant-tag of each invoice line is shown on p.7.)

4 · Sequencing — two hard dependencies + 90-day plan

- **No compute energizes before Quote 2 circuits.** The RTX workstation (575–600W) and DGX nodes need the dedicated 20A/30A circuits first.
- **The studio is unusable before Quote 3 drops.** Room 409 currently has *one* data drop; the 10GbE backbone + new drops come before deployment.

Phase	Actions
Now → **June 30** (grant window)	Lock education-priced POs for Quote 1 (quotes expire 14–30 days); commission the licensed-electrician load study; get the campus-IT vendor ruling (UniFi vs. Aruba/Cisco); confirm tax-exempt status + line-by-line grant eligibility.
30 days	Sealed sub bids (3 each: electrical, low-voltage, access, GC); permits; cabling + raised-floor design; finalize room dimensions & seat count.
90 days	Install **power** → **cabling/network** → **compute** → **cameras/access** → **demo day.** Display boards + dashboard go live; first cohort onboards.

5 · Existing-Conditions Assessment (Walkthrough, May 26 2026)

A true on-site assessment was performed with Dr. Smith. The conditions below — confirmed in the 16 verified photographs on p.4 — are what the three quotes resolve. Room → lab mapping:

Room	Becomes	Condition found (drives scope)
409	AI Studio (10–13 seats)	Dr. Smith's preferred room. **One data drop**; breakers trip on space heaters; keep the dual smartboards (one Wacom) + 2 projectors; fire-egress concern.
407	Math & Soldering Lab	Convertible; **empty built-in cabinetry** stores the soldering bench; an "L-bend" fits a fume extractor; shared CS-circuitry + engineering.
305 + 310	Engineering Lab	305 = project room w/ benches, packing tables, a donated 3D printer. 310 downstairs = the "good room" with **existing NVIDIA-GPU desktops to repurpose for AI hosting** (saves the engineering budget).
MDF closet	Network core	"Frankenstein" — switch *and* fuse box in one closet (not to code); drops out ~monthly; breakers too old. The leverage point to fix infrastructure.
405	(Vision only)	Borrow space for a small unisex bathroom — *facility/donor scope, excluded from the recommended tier.*

Whole-complex conditions: no 90° angles (slanted walls/floors); 15 "Frankensteined" AC units; all windows need resealing; bare cinderblock, no insulation. These raise facility labor and are why the load study + sealed bids precede commitment.

In Dr. Smith's words

> *"I've only got the one data drop right there." — (asked: in the entire room?) — "Yes, that's it right there."*

> *"These electrical units, they can't handle nothing… I got portable heaters… Boom." → "That means your breakers need to be updated."*

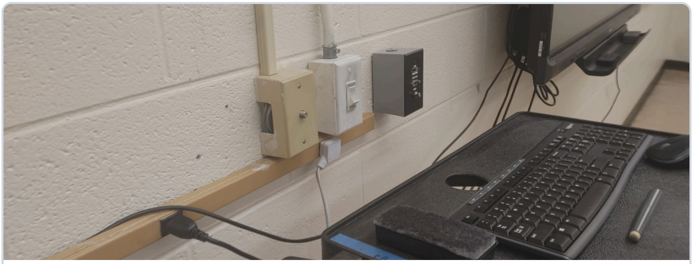
> *"I know this is not to code. We're not supposed to have a closet next to the fuse box."*

> *"The main objective: secure and always changing… nobody could copy it or retain that information." (access control)*

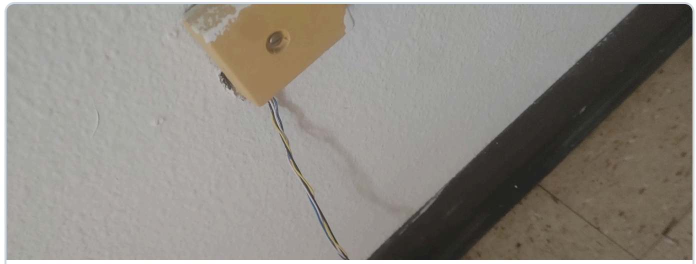
> *"We want AI. We need power. Power is the number one thing you invest in."*

6 · Photo Exhibit Addendum (verified on-site)

Each photograph was individually reviewed; captions describe what is actually in frame. Full-resolution files in `/exhibits`.



EX-01 · Rm 409: Teaching cart labeled "…MATH & COMP" fed by a **single surface-run outlet on bare cinderblock** — the core power/data need.



EX-03: Damaged low-voltage plate with **bare blue/yellow data wires hanging to the floor** — existing cabling is inadequate.

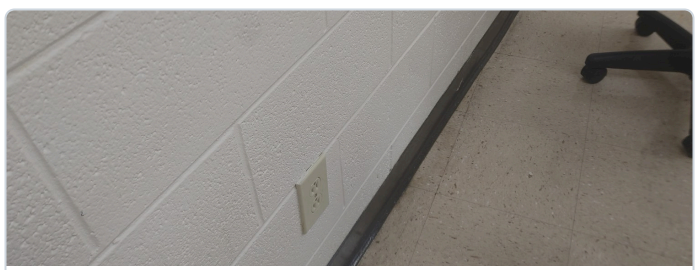
EX-09 · Rm 305: The department **already 3D-prints** (MakerBot-class) in a cluttered, unventilated corner — justifies the Bambu fleet + fume extraction.

EX-07: Reusable **blue engineering benches + pegboard**; this room already has a ceiling-cassette AC + projector. Trims furniture cost.

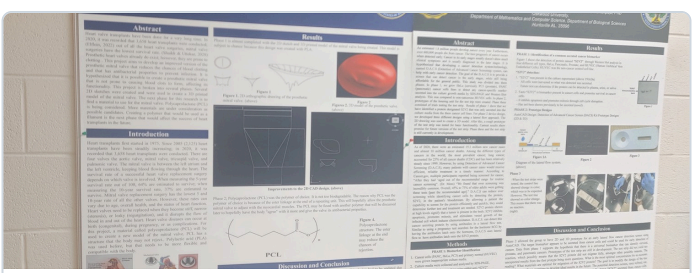
EX-05: Interactive panel on bare block with **exposed AV cabling**; a reusable blue cabinet at left.

EX-04: Dated 2-toggle ON/OFF control plate — the "confusing, unlabeled controls" finding.

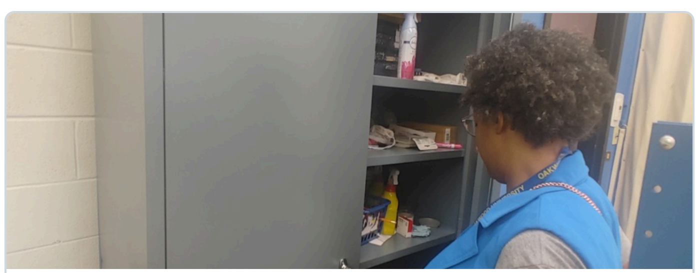
EX-06: Fogged / seal-failed window — envelope & HVAC weakness; supports supplemental cooling for GPU heat.



EX-02: A single low duplex outlet serving a classroom zone — extension-cord dependence.



EX-12: Real Oakwood student research (prosthetic mitral valve; cancer-screening "DACS") — the program already produces serious work.



EX-08: Reusable steel storage cabinet — supports the "separate machines from supplies / lockable storage" plan.

EX-14: Classroom door + lockset — the multi-door access-control + labeling retrofit.

EX-13: HVAC diffuser on bare block — part of the 15-unit "Frankensteined" system.

Additional verified exhibits in `/exhibits`: EX-10 lecture seating to reconfigure · EX-11 building access corridor · EX-15 "Graduates" pride board · EX-16 lab counter with a single outlet. 27 of 71 photos were blocked from download; re-share to add more.

7 · Quote 1 — AI Studio + Engineering Lab (Title III grant-eligible)

AI Studio · Rm 409

10–13 high-memory seats so students run local models: **6× Framework Desktop 128GB** (~$2,851) + **4× Mac Studio M4 Max 64GB** (~$2,600) + **1× DGX Spark** ($4,699) + an **RTX 5090 workstation** + **Jetson AGX Thor** + Orin Nano kits + Synology NAS + rack UPS + an **85″ demo wall**. Keep the dual smartboards/2 projectors.

Math & Soldering · Rm 407

5–10× Hakko FX-888DX soldering stations + recirculating **BOFA 3D PrintPRO**-class fume/VOC extraction in the cabinetry/L-bend; bench instruments; an enclosed **Carbide Nomad 3** CNC. Reuses the empty built-in cabinets.

Engineering · Rm 305 + 310

≥4 Bambu Lab printers: **H2D** large-format + **X1E** enterprise + **3× P1S + AMS**, enclosures, filament. **Repurpose the existing NVIDIA-GPU desktops** in 310 for AI hosting (10GbE NICs, RAM, IP-KVM) — engineering compute for near-zero new spend.

Division	Representative real SKUs (June 2026)	Planning range
AI Studio compute (lean)	Framework Desktop 128GB · Mac Studio M4 Max 64GB · DGX Spark · RTX 5090 WS · Jetson Thor/Orin · Synology DS1825+ · rack UPS	$30,000–$33,000
Engineering Bambu fleet + GPU repurposing	Bambu H2D AMS Combo ~$2,199 · X1E ~$2,649 · 3× P1S+AMS ~$799 · BOFA PrintPRO 3 · Intel X550-T2 NICs · PiKVM V4	$16,900–$18,500
Soldering bench + CNC	5–10× Hakko FX-888DX (~$135) · bench fume extraction · Carbide Nomad 3 ~$2,800	$8,000–$9,000
AV: 3 hallway display boards + 85″ studio demo wall	3× ~50″ commercial displays + UniFi Connect signage/media players + mounts · 85″ commercial demo wall ~$4,600	$8,250–$9,120
Furniture (movable) · Consumables/safety · Software/warranty	Movable benches/chairs · filament/resin/PPE/flammables cabinet/ABC+Class-D extinguishers · licenses + extended warranty	$9,800–$11,900
Quote 1 grant-eligible subtotal (equipment + supplies)		**$88,242–$98,820**

Display / project boards (your spec): a board outside *each* of the three labs (409 / 407 / 305-310), auto-fed from in-room activity via UniFi Connect — "you see AI happening before you enter the room." **Per-semester access** + the **85″ demo wall** make the studio look intentional and impressive. **Grant tag:** items ≥$2,000/unit are EQUIPMENT; the many <$2,000 items (Orin Nano, P1S, Hakko, cameras, NAS, UPS, switches) are SUPPLIES — confirm with the grants officer whether rack components aggregate to a capitalized system.

8 · Quote 2 — Power / Electrical (Facility · NOT grant · subbed + GC-managed)

Scope (lab-focused)	Planning range
100A lab subpanel · 14 dedicated 20A circuits · 2× 30A L5-30 UPS circuits · conduit/raceway on cinderblock · lab touch-ups · permits	$34,000–$37,000

Load calc (electrician verifies): ~7–9kW continuous / ~12kW peak (~60–75A @120V) → a dedicated 100A subpanel is the floor (NEC 125% continuous derating; resolve 120/240V vs 208V; size the CNC circuit for inrush). **Named Huntsville subs to bid:** Madison Electric, Pro Electric, Lee Company, Blake Brothers, Urban Electric.

9 · Quote 3 — Network + Wi-Fi + Cameras + Access (Mixed: electronics grant / cabling facility)

Division	Real SKUs (June 2026)	Range	Lane
UniFi core + pfSense	UniFi EnterpriseXG-24 ($1,299) · 2× USW-Pro-Max-24-PoE ($799) · UDM-Pro-Max ($599) · **Netgate 6100 MAX pfSense** ($999) · 42U rack	$3,837–$4,500	Grant (equip)
UniFi Protect cameras	UNVR-Pro ($499) + 8TB surveillance HDDs · 10–12× G5/G6 Turret/Bullet ($129–$199) · AI Pro ($499) for the print room	$3,055–$3,800	Grant (equip)
UniFi Access — 4 doors, rotating credentials	4–5× UA-Hub-Door ($199) · G3 Reader Pro ($379) at entry + G3 Readers ($159) · fail-safe HES strikes/maglock + REX + DPS + install	$8,400–$9,000	Equip + install
Structured cabling — lab-focused (SUBBED)	Cat6A drops + patch panel + MDF rack rebuild (cabling labor = facility)	$12,000–$14,000	Facility

Per-semester rotating credentials are native to UniFi Access (PIN expiry windows, NFC card deactivation, mobile/Wallet revoke — create a "Fall-2026" group, set term dates, deactivate at end → every credential dies at once). **Everything in one pane** (UniFi OS) replaces the antiquated MDF. **Named Huntsville subs:** low-voltage — Travel Tech (256.970.4558), Interweave (256-837-2300), Heart Technologies (256-859-4000), State Systems; access — AAC Systems, Clear Link, LOCKTEC (256-881-9300), Tri-State Electrical.

ADA / egress & code: all locks specified **fail-safe** (release on power loss); maglocks tie to the fire panel with REX + marked push-to-exit; Huntsville AHJ sign-off required. A distinct low-voltage permit + fire/egress coordination is carried in Permits.

10 · Execute-Ready Invoice — Recommended Execution

Invoice basis: IDC (Jonathan Isaac) as General Contractor. Grant-eligible lines may be procured under Title III; facility lines are subbed and carry GC coordination. Re-quote each line to a sealed bid before purchase.

#	Line item	Amount (range)
I · Grant-Eligible Equipment & Supplies (Title III)		
1	AI Studio compute — studio/lean tier	$30,000–$33,000
2	Engineering Bambu fleet + GPU-PC repurposing	$16,900–$18,500
3	Soldering bench + CNC	$8,000–$9,000
4	Consumables / safety / storage	$3,000–$3,800
5	AV — 3 hallway display boards + 85″ studio demo wall	$8,250–$9,120
6	UniFi network core + pfSense edge	$3,837–$4,500
7	UniFi Protect cameras — priority labs + key hallway	$3,055–$3,800
8	UniFi Access — 4 doors, per-semester rotating credentials	$8,400–$9,000
9	Furniture (movable lab/studio)	$5,000–$5,800
10	Software / licensing / extended warranty	$1,800–$2,300
Grant-eligible subtotal		**$88,242–$98,820**
II · Facility — Capital / Donor (NOT grant-eligible · SUBBED)		
11	Power / electrical — lab-focused (subpanel, circuits, raceway, permits)	$34,000–$37,000
12	Structured cabling — lab-focused (Cat6A drops, patch, MDF rebuild)	$12,000–$14,000
13	Raised access floor — AI Studio only (~300–400 sqft) + light room mod	$8,000–$10,000
14	Permits / insurance / freight	$2,400–$2,800
Facility subtotal		**$56,400–$63,800**
Direct cost (I + II)		**$144,642–$162,620**
III · GC Coordination, Contingency & IDC Fee		
15	GC coordination markup — 3–7% on the subbed facility base only	$1,692–$4,466
16	Contingency — 6–8% on (direct + markup); 2026 memory-price volatility	$8,780–$13,367
17	**IDC Orchestration Fee — 10% of pre-fee subtotal**	$15,511–$18,045
ALL-IN TOTAL · Recommended commitment $172,000		**$170,625–$198,498**

Payment schedule (proposed)

- **25% mobilization** — on signature; initiates procurement & sub scheduling.
- **50% progress** — at equipment delivery + electrical/cabling rough-in.
- **25% completion** — at commissioning, demo day, & closeout docs.

Pay deposit: IDC Stripe payment link (insert link)

Acceptance & signatures

Signature-ready; DocuSign / SignWell / Adobe / print-scan accepted.

Provider
Jonathan Isaac
Island Development Crew LLC
jon-isaac@islanddevcrew.com

Signature / Date

Client
Dr. Shushannah Smith
Oakwood University · Math & CS
ssmith@oakwood.edu

Signature / Date

11 · Curriculum Spine — *2084* / John Lennox (responsible-AI ethics for an Adventist HBCU)

Textbook: **"2084 and the AI Revolution"** (Updated & Expanded Ed., John C. Lennox, Zondervan Reflective 2024, ISBN 9780310166641, ~$24–28) — chosen over the 2020 original because it engages the generative-AI/LLM era. Companion: the *2084* 13-session Video Study for a flipped classroom. Two interleaved courses share one ethics spine; every 1–2 weeks ships one of **8 portfolio projects** ("students don't study AI — they ship it").

Wk	Ethics seminar (Lennox)	AI Engineering (Rm 409/310)	Project shipped
1	Ch.1 Mapping the territory	Onboarding; Python/Git; local vs cloud LLM	— setup
2	Ch.2–3 Origins / surveillance	RAG chatbot on a local model (Rm 310 GPUs)	**P1 Local RAG assistant**
3–4	Ch.4–5 Narrow AI: promise & bias	CV classifier + bias/fairness audit; model cards	**P2 CV model + bias audit**
5	Ch.6 Upgrading humans	Voice agent; assistive-tech use case + fabricate device	**P3 Assistive-tech AI device**
6–7	Ch.7–8 AGI risk / personhood	Agentic workflow + guardrails; multimodal app deploy	**P4 Agentic app w/ guardrails**
8–9	Ch.9 Origin of moral sense	Eval harness; red-team; responsible-AI write-up	**P5–P6 Eval + governance memo**
10–12	Ch.10–13 Homo Deus / Revelation	Capstone: ship a campus tool (à la Campus Companion)	**P7–P8 Capstone + demo day**

Pricing model

- **Digital Pass — $250–$350 / student:** lab access, the local-AI toolchain, project templates, the eval harness, demo-board publishing, and a portfolio site.
- **Textbook — ~$35:** *2084 and the AI Revolution* (bulk/ministry discounts via 10ofThose).
- Two new courses: **AI Engineering** + **hands-on Machining/CNC** — reduces required external co-op hours.

Why it lands at Oakwood

As the only SDA HBCU, Oakwood can teach AI with a **worldview-grounded ethics voice** secular schools struggle to articulate — data dignity, human-in-the-loop, Sabbath as a model for digital rest. Graduates leave with a portfolio *and* a point of view.

12 · Fundraising, Match Funds & the Naming Opportunity

Grant & corporate funding

- **Federal HBCU + AI:** NSF ExpandAI, NSF HBCU-EiR, NSF HBCU-UP, Dept of Ed Title III Part B (Oakwood is an active recipient).
- **Compute/hardware programs:** NVIDIA Academic Grant, AMD University Program, Google.org / UNCF.
- **Mission-fit philanthropy:** John Templeton Foundation ("Religion, Science & Society — Intelligence"), Lilly Endowment (faith & AI), Mellon ("Unruly Intelligences").
- **Corporate matching-gift** programs for employee donations.

Notoriety & the naming track

- **Voices to engage:** Dr. James Tour, John Lennox, Wesley Huff — credible on AI + faith + science.
- **Channels:** Modern Wisdom, Valuetainment, Diary of a CEO — to spotlight an HBCU building responsible AI.
- **Naming opportunity: "The John Lennox AI Innovation Lab"** (pending Oakwood approval) — pairs the *2084* curriculum with a named-gift envelope that funds the grant-ineligible facility scope.

13 · Department-Wide Vision (donor / naming ask · $245K–$330K)

The full robust transformation — the envelope to put in front of a donor or match funder, since it carries the facility scope Title III cannot.

Division (full scope)	Range	Division (full scope)	Range
AI Studio compute (pooled · RTX PRO 6000)	$42K–$50K	Power / electrical (full dept)	$34K–$48K
Engineering Bambu fleet (full)	$20K–$23.9K	Structured cabling (80 drops + fiber)	$24K–$36K
Soldering + CNC (full)	$11K–$14K	Raised floor + Room-405 bathroom + room mod	$45.6K–$62K
Cameras (dept-wide) + Access (5 doors)	$15K–$19K	Network core, AV, furniture, software, permits	$24K–$30K
Department-Wide Vision — all-in (incl. GC markup + 10% IDC fee)			**$245,000–$330,000**

14 · Orchestration, Reporting & Next Steps

Single point of accountability

IDC engages and schedules every vendor & Huntsville sub, manages the grant vs. facility funding lanes, and owns the timeline. One markup, one fee, one invoice.

Live progress & reporting

A live HTML **progress dashboard** link is attached to the project email; **weekly & monthly** progress reports keep faculty & administration abreast through commissioning.

Protected investment

Department cameras + per-semester rotating access control + cabling/power done to code mean the equipment is controlled, managed, and protected from day one.

Open items to confirm before sealed bids

- Final room dimensions & seat count (Rm 409 measured pass) — required for the raised-floor quote.
- Campus-IT vendor ruling: UniFi/pfSense allowed, or map to Aruba/Cisco/Extreme? Does access tie to the campus card?
- Oakwood tax-exempt status + grant eligibility ruling (setup labor, software/SaaS, networking hardware, cabling labor, access hardware).
- Tier to commit: Recommended ~$172K vs. a sub-$170K trim (single-lab cameras / 2-door access).
- Stripe Payment Link + DocuSign send authorization; Gmail authorization for automated progress emails.

Recommended next step. Approve the Recommended Execution tier (~$172,000 all-in), release the mobilization deposit, and authorize IDC to (1) commission the licensed-electrician load study and (2) issue the same scope to three sealed sub bids per trade. IDC returns firm, signed numbers within ~10 business days and begins the 90-day build.

Live reference hub: oakwood-ai-hub.vercel.app/ai-lab-final · Prepared by Island Development Crew LLC. Planning-grade; not a fixed bid until sealed sub quotes and the electrician load study are returned. Hardware prices are June-2026 and time-sensitive (2026 memory market) — re-quote within 14–30 days.

Authorized to proceed — Provider
Jonathan Isaac · Island Development Crew LLC

Signature / Date

Authorized to proceed — Client
Dr. Shushannah Smith · Oakwood University

Signature / Date